|INCH-POUND|

MIL-C-63292 (AR)
AMENDMENT 9
31 October 1995

SUPERSEDING
AMENDMENT 8
24 May 1993

MILITARY SPECIFICATION

CASE, CARTRIDGE, 105MM, M148A1B1

This amendment forms a part of Military Specification MIL-C-63292 (AR), dated 8 February 1980, and is approved for use by the U.S. Army Armament Research, Development and Engineering Center and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

* 2.1 STANDARDS MILITARY: Add:

MIL-STD-410 - Non-Destructive Testing Personnel Qualification and Certification."

DRAWINGS, ARMY: Add:

"9280433 - Envelope Drawing, 105MM Cartridge Case, Ultrasonic Inspection.

9280434 - Longitudinal Calibration Standards, 105MM Cartridge Case, Ultrasonic Inspection.

9280435 - Circumferential Calibration Standards, 105MM Cartridge Case, Ultrasonic Inspection."

Under AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI) Add:

* "ANSI/ASQCZ1.4 - Sampling Procedure and Tables for Inspection by Attributes."

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PAGE 3

* Add new paragraph 3.2.1:

"3.2.1 Cartridge cases. Cartridge case lots which are to be used in M900 cartridge lots must have satisfactorily passed a sticker test fired at M900 pressures (see 4.5.6.3) in addition to meeting the requirements of this specification. No replating is permitted on cases for the M900 cartridge."

PAGE 4

3.8.1: After "free from" add "...dents, shiny spots, nicks or gouges, wrinkles...".

3.8.4: Delete "No cartridge case shall have a burr." and substitute "Burrs shall be limited to those described in 105mm cartridge case 'Visual Defect Standards'."

PAGE 5

4.3.1:

Delete "254*" and substitute "262".

Delete "* All samples must meet the requirements for thermal treatment and high stress expansion test as performed in the uncoated state."

PAGE 6

Table I: First article inspection.

Delete "Cartridge Case - Prior to Coating".

Delete "Cartridge Case - After Coating".

Delete "Examination for defects 20 3.1 4.4.2.1".

Add "Ultrasonic inspection 262 3.8.1 4.5.7.1".

Examination for defects, delete "254" and substitute "262".

PAGE 7

* 4.4.2.a, delete "MIL-STD-105" and substitute "ANSI/ASQCZ1.4".

PAGE 8

4.4.2.1: Delete in its entirety and substitute included 4.4.2.1.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Case, Cartridge - Prior to Coating	1 OF 1	10522799
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	High stress expansion test		100%(a)	3.1	4.5.2

NOTES: (a) All non-compliance units must be removed from the lot. Evidence of cracks in any portion of the case shall be cause for rejection.

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4.4.2.2: Delete in its entirety and substitute included 4.4.2.2.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating	1 OF 3	10522799
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cracks, metal and plating defects		100%(a)	3.4 & 3.8.1	4.5.7
102	Deleted				
103	Thickness of coating (See 4.5.6)			3.4	4.5.6.1
104	Salt spray (See 4.5.6)			3.1	4.5.6.2
105	Maximum profile and alignment		100%(a)	3.5	4.5.8
106	Excess pressure ballistic test	(d)		3.6	4.5.9.2.1
107	True position of pitch diameter of thread with body		0.40%	3.1	Gage
108	True position of primer hole counterbore with pitch diameter of thread		0.40%	3.1	Gage
109	True position of primer hole thread relief with pitch diameter of thread		0.40%	3.1	Gage
110	Pitch diameter of thread, max.		0.40%	3.1	Gage
111	Minor diameter of thread, max.		0.40%	3.1	Gage
112	Length total, max.		0.40%	3.1	Gage
113	Diameter of primer hole counterbore		0.40%	3.1	Gage
114	Depth of primer hole counterbore, min.		0.40%	3.1	Gage
115	Length from flange to bottom of primer hole counterbore, max.		0.40%	3.1	Gage
116	Depth of thread relief		0.40%	3.1	Gage
117	Deleted				

NOTES:
(a) All non-compliance units must be removed from the lot.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating		10522799
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Major					
118	Ultrasonic acceptance stamp missing		100%	3.8.1	4.5.7.1
119	Hardness	4	(b)	3.3	4.5.3
120	Macro-examination	4	(b)	3.8.2	4.5.5
121	Diameter of flange, min.		0.40%	3.1	Gage
122	Diameter of mouth		0.40%	3.1	Gage
123	Deleted				
124	Thickness of flange, min.		0.40%	3.1	Gage
125	Thickness of base, min.		0.40%	3.1	Gage
126	Wall thickness at first datum from base		0.40%	3.1	Gage
127	Wall thickness at second datum from base		0.40%	3.1	Gage
128	Wall thickness at third datum from base		0.40%	3.1	Gage
129	Wall thickness at mouth		0.40%	3.1	Gage
130	Contour of inside of base and sidewall not smooth		0.40%	3.1	Visual
131	Ultrasonic inspection		100%	3.8.1	4.5.7.1
132	Ultrasonic inspection stamp missing		100%	3.8.1	4.5.7.1
Minor					
201	Pitch diameter of thread, min.		0.65%	3.1	Gage
202	Diameter of primer hole thread relief		0.65%	3.1	Gage
203	Thread damaged or not full		0.65%	3.8.3	Visual
204	Deleted				
205	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:
(b) Reject on one defect.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Case, Cartridge - After Coating	3 OF 3	10522799
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
206	Wax coating missing		0.65%	3.1	Visual
207	Evidence of poor workmanship		0.65%	3.8.4 & 3.8.5	Visual
208	Diameter of body under flange, min.		0.65%	3.1	Gage
209	Location of first basic diameter from flange, min.		0.65%	3.1	Gage
210	Location of second basic diameter from flange, min.		0.65%	3.1	Gage
211	Location of third basic diameter from flange, min.		0.65%	3.1	Gage
212	Location of fourth basic diameter from flange, min.		0.65%	3.1	Gage
213	Radii or chamfers missing or incorrect		0.65%	3.1	Visual
214	Surface finish improper		0.65%	3.1	Visual
215	Length total, min.		0.65%	3.1	Gage
Special					
a.	Tension test (information only)	(c)		3.3	4.5.4
b.	Excess pressure ballistic test split or crack within 12 inches of base	(d)	4.4.3.1.2	3.6	4.5.9.2.1

NOTES: (c) Samples used for the hardness test shall be used in the tension test (see 4.5.4).
(d) Major 106 and special b. utilize the same sample. Ref. 4.4.3.1.2.

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4.4.2.3: Major defect: Delete "None defined" and add:

"101 Cracks and other metal defects (cartridge cases) just prior to packing into carton 100% 3.8.1 Visual".

PAGE 14

4.4.3.1.1:

Table II, Defect characteristic 2: Delete: "one caliber of the head" and substitute "12 inches of the base".

Table II, Defect characteristic 3: Delete: "one caliber from the head" and substitute "12 inches from the base".

4.4.3.1.2

Table III, Defect characteristic 2: Delete: "one caliber of the head" and substitute "12 inches of the base".

Table III, Defect characteristic 3: Delete: "one caliber from the head" and substitute "12 inches from the base".

After "Production lot" add "(Cases not ultrasonically tested)".

Line 6, delete: "...provided for the test." and substitute "...tested."

Add paragraph 4.4.3.1.3:

"4.4.3.1.3 Production lot (Cases ultrasonically tested). Twenty five cartridge cases from each lot (see 4.4.1), shall be ballistically tested in the excess pressure phase. Results of the test shall be evaluated in accordance with the production acceptance criteria set forth in Table III. Except for all defect characteristics, the acceptance criteria are accept on 0 defects, reject on 1 defect."

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4.5.4: Delete in its entirety and substitute:

"4.5.4 Tension test (information only). Four cartridge cases from each lot shall be subjected to this test. The test specimens shall be taken from each cartridge case in accordance with the standard position and size of the tensile specimen chart described in Figure 1. The yield strength, tensile strength and elongation shall be determined in accordance with the procedures prescribed by ASTM-E8. The results shall be recorded for information only."

PAGE 16

* 4.5.6.1: Last sentence, delete "...rejection of the lot." and substitute "...testing additional samples in accordance with ANSI/ASQCZ1.4, Inspection Level II, 0.40% AQL."

* Add new paragraphs 4.5.6.3 and 4.5.6.4:

"4.5.6.3 Cartridge case sticker test. Ten (10) M900 slugs shall be assembled into cartridge cases which have been loaded with M43 propellant to achieve a resulting pressure of 86,600 pounds per square inch (psi) (the permissible maximum pressure (PMP)) when fired at 120°F. The cartridge shall be stored in a temperature conditioning unit (TCU) which is maintained at 120°F - 5°F for a minimum of 40 hours prior to firing from a ballistically accepted gun tube. The fired cartridge cases must meet the requirements for obturation, soundness, extraction and primer retention listed in this specification. This test shall be performed once for each cartridge case lot used in the assembly of M900 cartridges. Three (3) additional cartridge case samples shall be taken from every lot for replacement of disregards if needed. Uploading of the test rounds during the course of the test should be performed in order to continually achieve 86,600 psi for all test rounds.

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4.5.6.4 Test evaluation method. Any round that achieves a pressure greater than 86,600 psi and fails the requirements of this specification shall be classified as a disregard. Any round that achieves a pressure less than 84,000 psi shall also be classified as a disregard unless it fails the requirement of this specification in which case it shall be classified as a failure. Any round that achieves a pressure in the range of 84,000 psi to 86,600 psi inclusive shall be considered acceptable unless it fails the requirements of this specification, in which case it shall be classified as a failure. All rounds that are considered as a disregard shall be replaced.

If any round is classified as a failure, then the lot of cases represented by these samples shall not be used for assembly of the M900 cartridge. In the event that there are fewer than 10 valid test points, the lot shall be referred to the government's technical agency for disposition."

4.5.7: Delete and substitute:

"4.5.7 Cartridge case cracks and flaw detection. The cartridge case shall be visually examined for cracks, defects and workmanship and shall comply with the requirements of paragraph 3.8 or established visual standards. The visual standards shall be mutually agreed upon between the government and the contractor. All rejected cases shall be set aside and held for government inspection."

4.5.7.1: Delete and substitute:

"4.5.7.1 Ultrasonic inspection. Each cartridge case shall be inspected for metal defects using as reject standards drawings 9280434 and 9280435 and equipment satisfying the requirements of drawing 9280433. Deviation from standards must be approved by the respective technical agency. All cartridge cases accepted by the ultrasonic equipment shall be stamped with a "U". Cartridge cases rejected by the equipment shall be stamped with an "R" and the rejected cases subsequently removed. The stamps utilized shall not have sharp corners which will create corrosion. The stamping shall comply with the requirements as stated in cartridge case drawing 10522799. The quantities of cartridge cases inspected and rejected shall be counted and logged for determination of the reject rate."

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Add paragraphs 4.5.7.1.1, 4.5.7.1.2, 4.5.7.1.3 and 4.5.7.2:

"4.5.7.1.1 Calibration procedures. The ultrasonic inspection equipment, specified by drawing 9280433, shall be calibrated, at a minimum, at the beginning, middle and end of an eight hour work shift. If for any reason the reject standards are not rejected by any of the channels, the contractor must re-calibrate the equipment and re-inspect all accepted cartridge cases ultrasonically inspected since the last acceptable calibration.

4.5.7.1.2 Rejection rate - ultrasonic equipment. When the ultrasonic inspection equipment rejection rate exceeds 5.0 percent for the period between calibrations, the contractor shall check equipment calibration. If the calibration is acceptable, all cartridge cases previously accepted by the equipment during this period shall be re-inspected. During re-inspection, if the reject rate exceeds 1.0 percent, all cartridge cases scheduled for re-inspection or accepted by re-inspection shall be placed aside until the cause for the excessive reject rate is determined. When the cause is determined, disposition of the held cartridge cases shall be at the discretion of the contracting officer. At any time, if the contractor proves that an excessive reject rate is equipment related, he may apply to the contracting officer for re-inspection of those rejected cartridge cases found during this time.

4.5.7.1.3 Ultrasonic equipment - operator qualification. The operators of the contractor's equipment for ultrasonic inspections required by this specification shall be certified and qualified "Level I". Also, any set-up, calibration or interpretation personnel shall be certified and qualified "Level II". The qualification and certification criteria shall be as prescribed by MIL-STD-410 and other applicable specifications.

4.5.7.2 Visual inspection. Four power (4x) magnification shall be used in the examination of the exterior surface of the sidewall for a minimum distance of twelve inches forward of the base (flange). All other surfaces may be examined without magnification."

4.5.8 First sentence: Delete "...8663760..." and substitute "...8647040...".

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4.5.9.1.3:

Second sentence: Delete "...product..." and substitute "...produce...".

Between "(498.49 MPa)" and "74,200 psi". add "...to...".

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